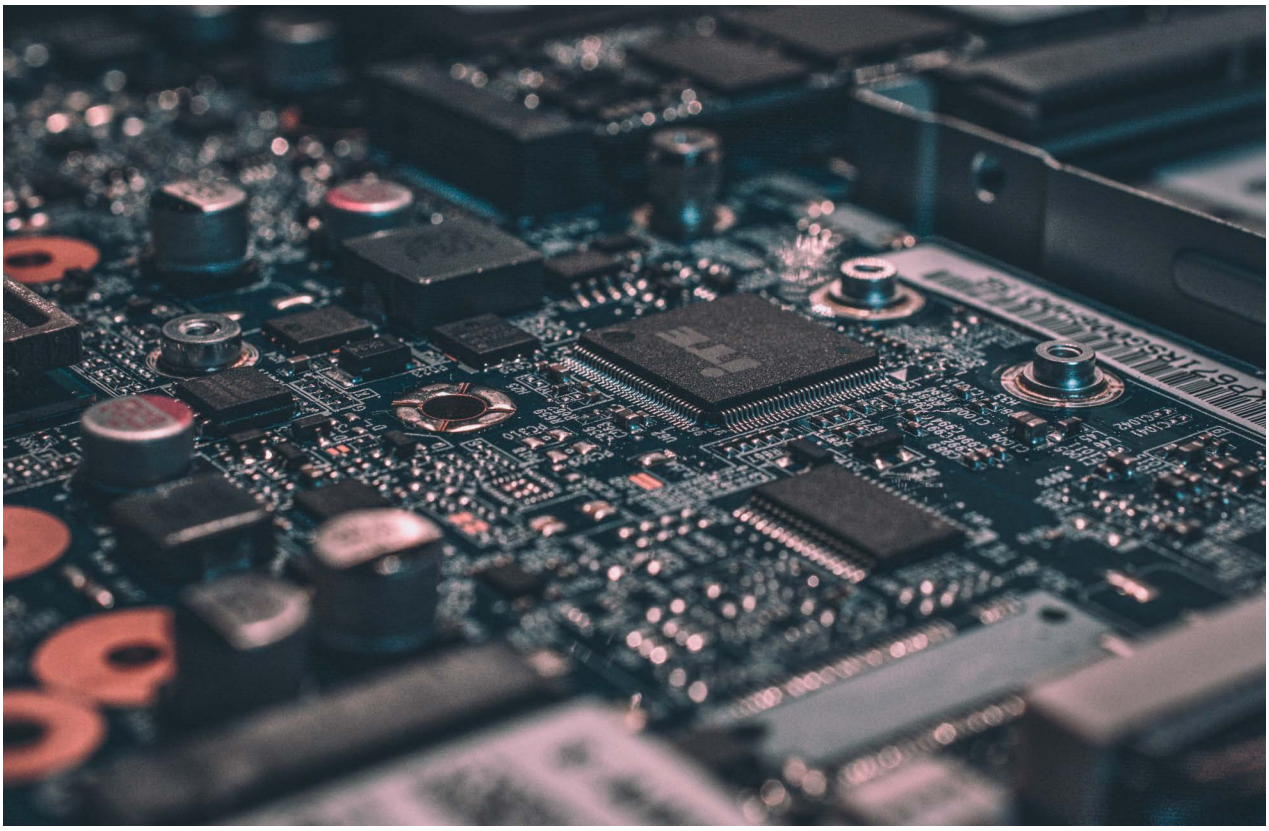


WaveDrom Starter Guide

WaveDrom is an open-source JavaScript application that generates digital timing diagrams inside a web browser. This allows you to create your own diagrams without needing to download any software.

This guide provides:

- Common applications of timing diagrams
- An overview of timing diagrams
- A step-by-step tutorial on how to create you own diagram using WaveDrom

What is a Timing Diagram?

A digital timing diagram is a visual representation of a set of digital signals over time.

Common uses for timing diagrams are:

- Electrical engineering
- Computer architecture
- Software debugging

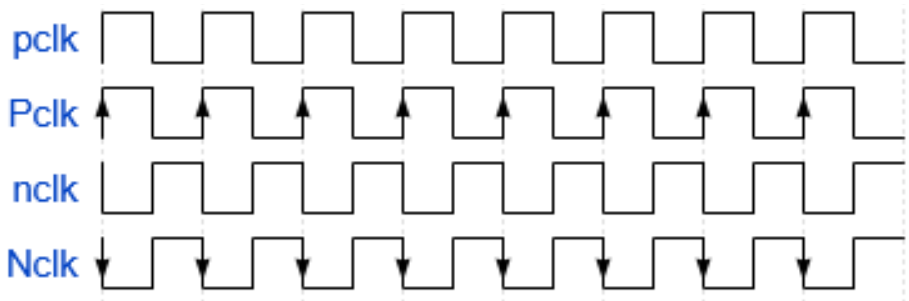
Components of a Timing Diagram

Before you launch WaveDrom, ensure you are familiar with the components of the diagram.

Clock Signal (CLK)

The clock acts as the heartbeat of a digital system, telling the circuit when to perform each operation. The clock alternates between high and low and can have a positive or negative polarity.

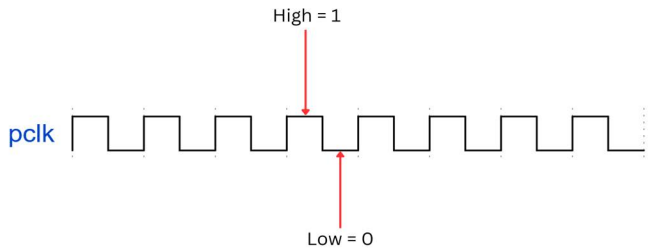
Example:



High and Low States

High and Low represent the two states of a digital signal. High states is the raised horizontal line and represents a one or “on” state. Low is the lowered horizontal line that represents zero or “off”

Example:

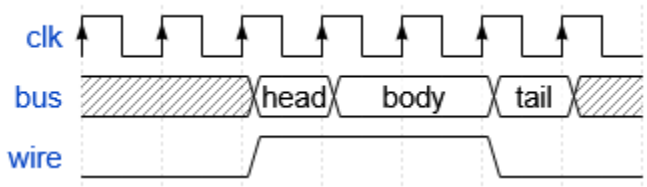


Data Buses

A data bus contains information that is being read from or written to the system.
On a timing diagram, data buses are represented by:

- Parallel Hexagonal Shapes
- A Single Center Line

Example:



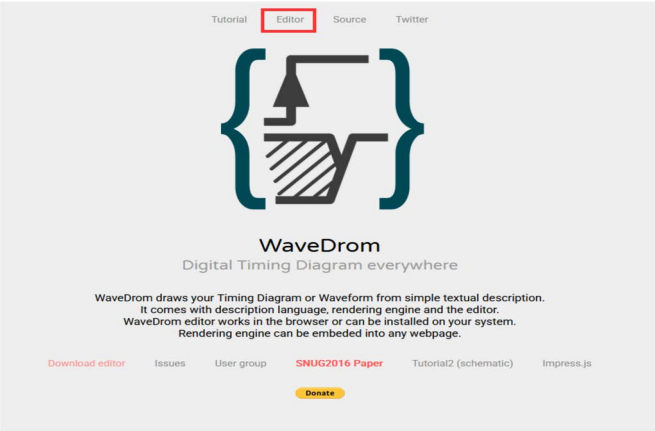
Creating a Diagram with WaveDrom

WaveDrom enables you to create your own timing diagrams and export them as image files directly from your browser.

To create a timing diagram, follow these steps:

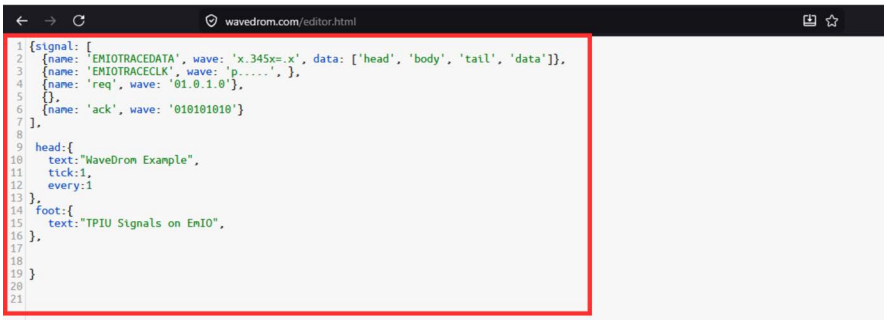
Note: If you are new to programming, taking a moment to learn basic JSON syntax will help you get the most out of WaveDrom.

1. Navigate to the WaveDrom website.
2. Select **Editor**. The editor page opens.



Note: For a details on syntax and a full list of WaveDrom's capabilities, see the tutorial page on the WaveDrom website.

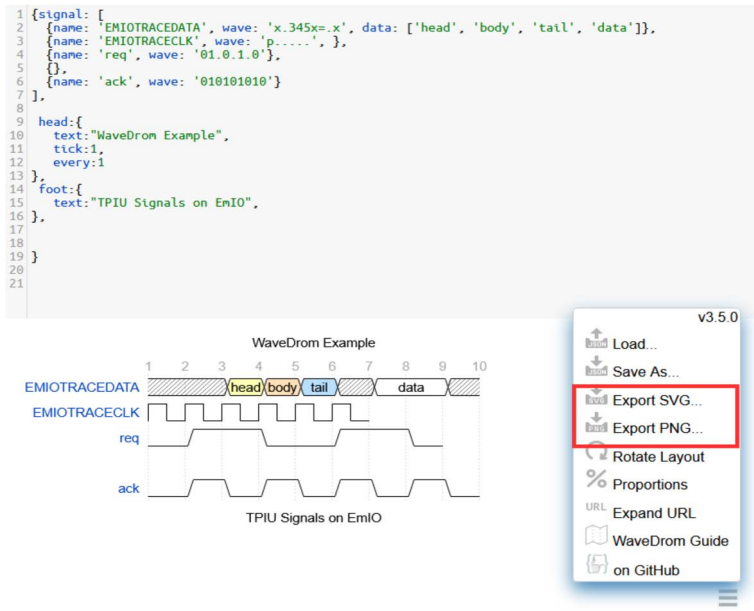
3. Modify the code as needed. The timing diagram at the bottom of the page updates in real time.

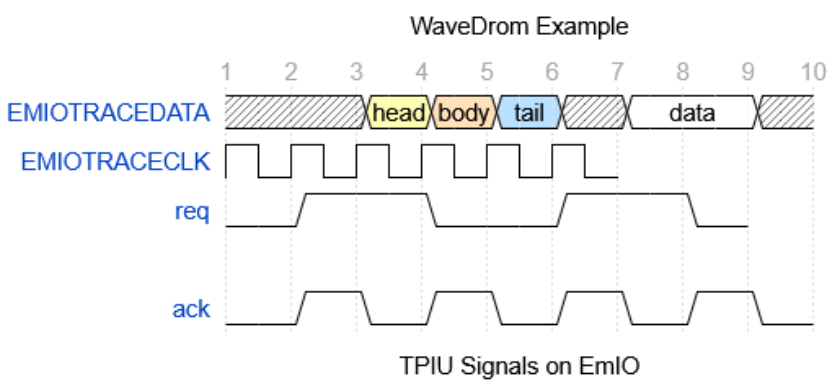


Exporting Your Diagram

To export your timing diagram, follow these steps:

1. In the bottom right of the page, select the button. A list of options appears.
2. Select an **Export** option. The file explorer opens.
   a. Select **Export SVG** for a vector image file.
   b. Select **Export PNG** for a raster image file.



3. Save the file to the desired location.

Your digital timing diagram is now in a shareable format.